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Multi-Horizon Time-Series Forecasting for Enterprise Analytics

(Balancing Simplicity, Real-World Impact, and Research Support)

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ABSTRACT

Accurate demand forecasting helps businesses optimize inventory, staffing, and resource planning. This project proposes a Multi-Horizon Time-Series Forecasting system that predicts multiple future time steps using ARIMA, Prophet, Temporal Fusion Transformer (TFT), and DeepAR. The system incorporates seasonality and external factors such as holidays and promotions, with data preprocessing, feature engineering, model training, and backtesting. Performance is evaluated using MAPE and RMSE to improve forecasting accuracy and planning efficiency

Problem Statement

- Enterprises rely heavily on accurate forecasting to make critical decisions related to inventory, staffing, and revenue planning. However, most existing forecasting systems predict only a single future time step, making them ineffective for simultaneous short-term operational planning and long-term strategic decision-making.

- Traditional statistical models also struggle to capture complex seasonal patterns, sudden demand shifts, and external influencing factors such as holidays and promotions, resulting in inconsistent and unreliable predictions across different planning horizons. This creates a need for a unified forecasting system capable of generating accurate, multi-horizon predictions to support better business decision-making.

INTRODUCTION

- Every day, businesses collect huge amounts of data — sales, orders, website visits, and stock levels. Since this data changes over time, it's called time-series data. Predicting what will happen next using this data is called forecasting, and it helps businesses plan ahead instead of reacting late.
- Most forecasting tools today only predict one point in the future, like tomorrow's sales. But real businesses need predictions for many time periods at once — tomorrow, next month, and next quarter — and all these should match each other, not be separate guesses.
- This project builds a system that predicts short-term, medium-term, and long-term results all at once, using a single model, so the predictions stay consistent.

- Earlier, businesses used simple methods like ARIMA, but these struggle with complex patterns and sudden changes like holidays or discounts. Later, machine learning models like XGBoost became popular for handling more complex data.
- Now, deep learning models like DeepAR and Temporal Fusion Transformer (TFT) are used because they understand complicated patterns better and can forecast many products at once.
- A key idea in this project is multi-horizon forecasting — instead of separate models for each time period, we use one model to predict all periods together. This is faster and keeps results consistent. But a real system also needs to collect data, retrain itself, monitor accuracy, and show results on simple dashboards.

LITERATURE REVIEW

S.No	Title of Paper	Year	Authors	Methodology / Algorithm	Objective	Limitation	How Proposed Work Addresses It
1	A Multiscale Model for Multivariate Time Series Forecasting	2025	Vahid Naghashi, Mounir Boukadoum, Abdoulaye Banire Diallo	Multiscale Transformer-based forecasting	Capture patterns at different temporal scales and inter-series relationships.	Primarily focuses on a multiscale deep-learning architecture rather than practical model selection.	Compare simple and complex models using multi-horizon backtesting and computational cost.
2	Adaptive Demand Forecasting Framework with Weighted Ensemble of Regression and Machine Learning Models Along Life Cycle Variability	2025	Islam M. Hammam, Amin K. El-Kharbotly, Yomna M. Sadek	ARIMA-based Regression + XGBoost + Weighted Average Ensemble	Forecast demand under different product life-cycle patterns.	Focuses on ARIMA/XGBoost ensemble and product life cycle; broader multi-horizon model comparison is limited.	Compare ARIMA/Prophet, LightGBM and optional N-BEATS/TFT across different horizons and evaluate business impact.
3	Improving Long-Term Multivariate Time Series Forecasting with a Seasonal-Trend Decomposition-Based 2D Temporal Convolution Dense Network	2024	Jianhua Hao, Fangai Liu	Seasonal-Trend Decomposition + 2D Temporal Convolution + Dense Network	Improve long-term multivariate forecasting by capturing temporal patterns.	Complex DL architecture; limited focus on simplicity and deployment.	Balance accuracy, interpretability and computational cost, with deployment considerations.

S.No	Title of Paper	Year	Authors	Methodology / Algorithm	Objective	Limitation	How Proposed Work Addresses It
4	Learning Evolving Relations for Multivariate Time Series Forecasting	2024	Binh Nguyen-Thai et al.	ERAN + Evolving Graph + Temporal Attention Convolution	Capture evolving inter- and intra-variable relationships.	Advanced architecture; limited practical model-selection perspective.	Select models based on horizon-wise performance and business requirements.
5	A Demand Forecasting System of Product Categories Defined by Their Time Series Using a Hybrid Approach of Ensemble Learning with Feature Engineering	2024	Santiago Mejía, Jose Aguilar	Time-series feature engineering + clustering + ML forecasting + ensemble learning	Improve retail product-demand forecasting by grouping products with similar demand behavior.	Focuses mainly on retail demand and ensemble ML; limited comparison with advanced DL and multiple enterprise horizons.	Extend to multi-horizon enterprise forecasting using statistical, ML and DL models.
6	Fuzzy Inference-Based LSTM for Long-Term Time Series Prediction	2023	Weina Wang et al.	Fuzzy Inference + LSTM	Improve long-term forecasting accuracy and interpretability.	Mainly LSTM-focused; limited enterprise/business evaluation.	Add model comparison + business-impact evaluation.
7	A Novel Encoder-Decoder Model for Multivariate Time Series Forecasting	2022	Huihui Zhang et al.	GRU Encoder-Decoder, Attention, Residual connections	Improve multivariate forecasting.	Single advanced architecture; limited business-oriented comparison.	Compare statistical + ML + DL models across multiple horizons.

Data Source

The M5 Forecasting Accuracy dataset is a large-scale retail demand forecasting dataset released by Walmart, containing historical daily unit sales from 10 stores across California, Texas, and Wisconsin. It includes thousands of product-store level time series, making it well suited for studying demand behavior across multiple forecasting horizons, from short-term operational planning to long-term strategic decisions. Beyond raw sales, the dataset provides calendar information such as dates, weekdays, months, and special events, along with selling-price data that acts as an external factor influencing demand.

- **M5 Forecasting – Accuracy (Kaggle) –** 42,840 hierarchical product-store series across 10 Walmart stores in CA, TX & WI; daily sales + calendar + price data (primary dataset)
- **Rossmann Store Sales (Kaggle) –** 1,115 German drugstores, ~3 years of daily sales with promo & holiday flags (cross-domain validation)
- **UCI Electricity Load Diagrams –** 370 clients, 15-min interval consumption over 4 years (high-frequency, long-horizon robustness check)
- **Walmart Recruiting – Store Sales Forecasting (Kaggle) –** 45 stores, weekly sales with markdown & holiday events (secondary validation set)
- **Why M5:** only dataset with product-store granularity across multiple regions needed to train and backtest

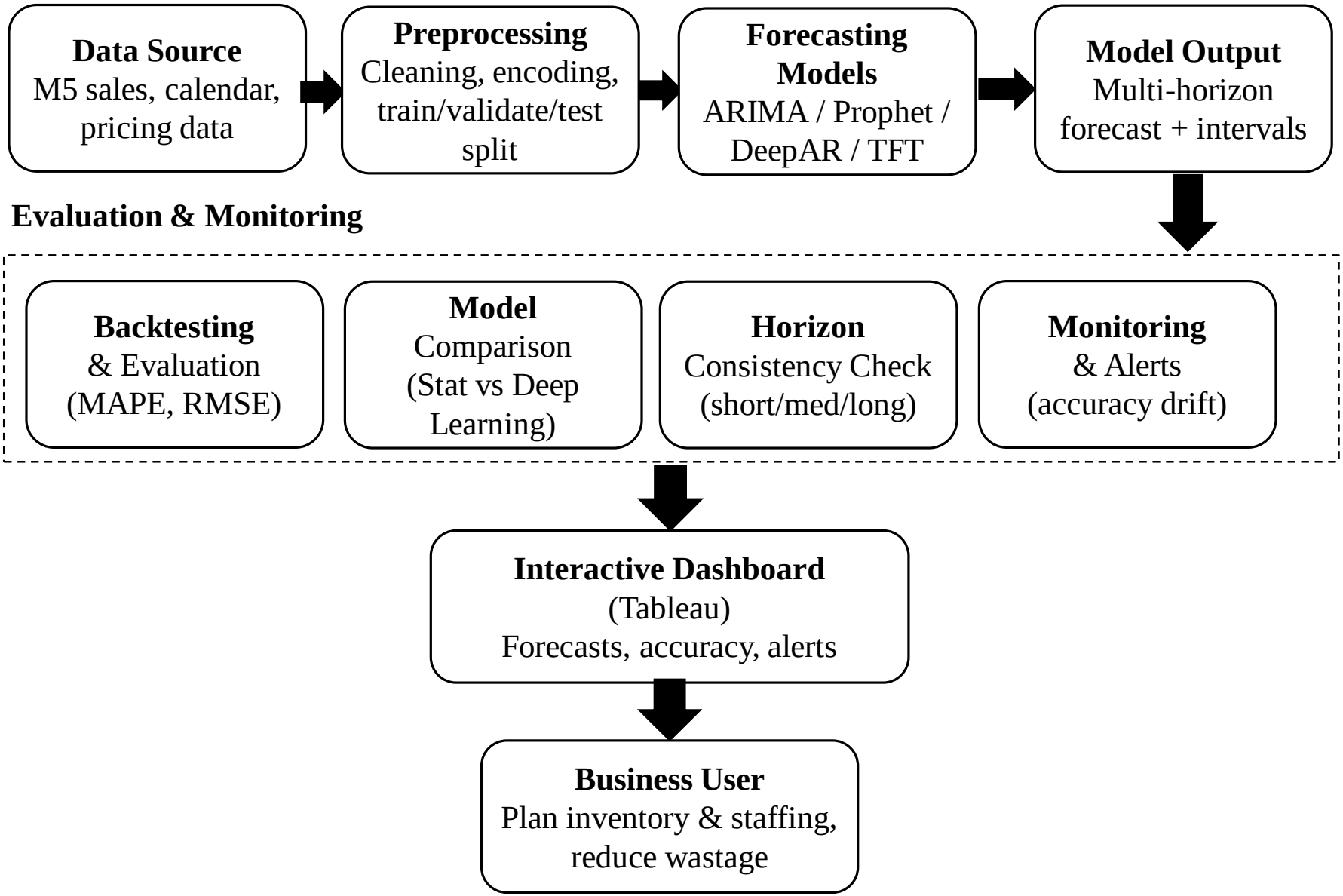
Links:

- **M5 Forecasting Accuracy –** https://www.kaggle.com/competitions/m5-forecasting-accuracy
- **Rossmann Store Sales –** https://www.kaggle.com/competitions/rossmann-store-sales
- **UCI Electricity Load Diagrams –** https://archive.ics.uci.edu/dataset/321/electricityloaddiagrams20112014
- **Walmart Recruiting Store Sales Forecasting –** https://www.kaggle.com/competitions/walmart-recruiting-store-sales-forecasting

Methodology

- **Domain Knowledge**
  - Enterprise multi-horizon demand forecasting
  - Focus on accuracy, consistency across horizons, and real-world usability
- **Model Development**
  - Preprocess the dataset (cleaning, calendar & lag features)
  - Train ARIMA/Prophet and DeepAR/TFT for multi-horizon forecasting
- **Evaluation**
  - Back test every model across short-, medium-, and long-term horizons
  - Compare performance using MAPE and RMSE
- **Monitoring & Deployment**
  - Track forecast accuracy over time for drift
  - Trigger automatic retraining alerts when accuracy drops
- **Decision Support**
  - Integrate all results into an interactive Tableau dashboard
  - Support inventory and staffing decisions with clear, trustworthy forecasts

System Architecture



Thank You